URL Shortener

Python Internship Project

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Project category: Python Programming

Repository: upskillcampus

PROJECT REPORT

1. Project Overview

A small command-line URL shortener that generates short codes for long URLs, stores mappings permanently in SQLite, and retrieves the original URL using a code.

2. Objectives

The objective is to build a practical Python application that demonstrates core programming concepts and provides a simple, usable solution to the selected task.

3. Main Features

• Accepts a long URL and creates a unique short code

• Stores URL mappings permanently in an SQLite database

• Looks up the original URL by short code

• Lists saved mappings and supports a quit command

4. Technologies Used

Python 3; SQLite; secrets; string; URL validation

5. Program Workflow

1. Start the Python program.
2. Display the menu or request the required input.
3. Validate input and perform the selected operation.
4. Display the result and handle common errors.
5. Save or retrieve data where the project requires persistent storage.

6. Testing

The program should be tested with normal inputs, invalid inputs, empty fields, and repeated runs. For file operations, use a sample directory first. For database-backed projects, confirm that saved data remains available after restarting the program.

7. Challenges and Solutions

Typical challenges include validating user input, handling duplicate records or filenames, and keeping data available across sessions. These are addressed with conditional checks, unique identifiers or destination names, and local database storage where applicable.

8. Learning Outcomes

This project provides practice with Python functions, control flow, data structures, input validation, file handling, and/or database operations. It also develops debugging and project documentation skills.

9. Source Code and Repository

Repository name: upskillcampus

GitHub profile: https://github.com/divagar5943-lang

Repository URL: https://github.com/divagar5943-lang/upskillcampus

Note: Upload the matching .py file and this PDF to the repository before submitting. The repository and file links must be checked after upload; this report does not claim that the files have already been uploaded.

Implementation Note

This version stores short codes in SQLite and can look up original URLs. It is a console demonstration; it does not create a publicly accessible HTTP redirect service.